THE HOMOGENEOUS IDEAL GENERATED BY THE HOMOGENEOUS ELEMENTS OF A PRIME IS PRIME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the frozen snapshot stages the chapter's own preamble file beside it (preamble.tex, 217 lines, 8116 bytes, the same shared Stacks-wide preamble the published chow and etale-cohomology cards use); the first line of the pinned chapter file is \input{preamble}, so no preamble text lives inside the chapter itself. The card takes that file's lines 8-18 (the verbatim package and the three \comment-based environments reference/slogan/history at lines 9-12, section 13.2, together with the Xy-pic setup) and its lines 154 to the runtime end of the file (the native theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- Primality tested on homogeneous elements. Used at the start of the counted proof (lines 13877-13878). Reduces the primality of the homogeneous ideal q to the case of a product of two homogeneous elements.
- Membership of a homogeneous element in a homogeneous ideal. Used at the end of the counted proof (lines 13882-13883). Upgrades membership in the prime p to membership in the homogeneous ideal q, which is generated by the homogeneous elements of p.

Lemma 0.1. *Let S be a graded ring. Let $\mathfrak{p} \subset S$ be a prime. Let $\mathfrak{q}$ be the homogeneous ideal of S generated by the homogeneous elements of $\mathfrak{p}$. Then $\mathfrak{q}$ is a prime ideal of S.*

Proof. To prove that $\mathfrak{q}$ is prime, it suffices to check that if $f, g \in S$ are homogeneous and $fg \in \mathfrak{q}$, then either f or g in $\mathfrak{q}$. Then $fg \in \mathfrak{p}$ because $\mathfrak{p} \subset \mathfrak{q}$. Since $\mathfrak{p}$ is prime we see that either $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$. Since f and g are homogeneous, it then is clear that either $f \in \mathfrak{q}$ or $g \in \mathfrak{q}$. □

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